

# Administrator Guide

WebFlowMaster Enterprise Documentation

Language: EN | Version 1.0.0

Generated on: 5/5/2026





# WebFlowMaster: Enterprise Administrator Guide

---

Governance, Security, and Infrastructure Maintenance



## Document Objective

---

This guide is intended for System Administrators, DevOps Engineers, and IT Managers. It covers advanced installation, user management, data security, and integration with corporate workflows.

---



## Table of Contents

---

1. [System Architecture](#)
  2. [Governance and RBAC \(Role-Based Access Control\)](#)
  3. [Security and Data Protection](#)
    - [AES-256 Encryption](#)
    - [Secret and API Key Management](#)
  4. [Worker Configuration and Scalability](#)
  5. [Enterprise Integrations \(CI/CD, Webhooks\)](#)
  6. [Maintenance and Disaster Recovery](#)
- 

## 1. 🏗️ System Architecture

---

WebFlowMaster is based on a micro-services architecture:

- **Core Server:** API management, business logic, and orchestration.
  - **Worker Engine:** Isolated instances for test execution (based on Playwright/Puppeteer).
  - **Persistence Layer:** SQL database for metadata and Redis for message queues (BullMQ).
  - **Storage Layer:** Dedicated system for storing screenshots and video traces.
- 

## 2. 👥 Governance and RBAC (Role-Based Access Control)

---

Access management is granular. Every user belongs to a role with predefined permissions:

| Permission       | Super Admin | Manager | Tester | Viewer |
|------------------|-------------|---------|--------|--------|
| User Management  | ✓           | ✗       | ✗      | ✗      |
| System Config    | ✓           | ✗       | ✗      | ✗      |
| Project Creation | ✓           | ✓       | ✗      | ✗      |
| Test Creation    | ✓           | ✓       | ✓      | ✗      |
| Test Execution   | ✓           | ✓       | ✓      | ✗      |
| Report Viewing   | ✓           | ✓       | ✓      | ✓      |

**Best Practice:** Use the `Viewer` role for stakeholders who only need to monitor results without risking changes to the test suites.

### 3. Security and Data Protection

#### A. AES-256 Encryption

All credentials saved in the platform (e.g., user passwords for test logins) are encrypted at rest using the AES-256-GCM standard. The master encryption key must be managed via protected environment variables.

#### B. Secret and API Key Management

API Keys generated for CI/CD integration must be rotated every 90 days. WebFlowMaster supports automatic masking of secrets in execution logs to prevent accidental data leaks.

### 4. Worker Configuration and Scalability

Workers are the computational engine. In high-load environments, you can scale horizontally by adding more Workers.

- **Concurrency:** Each worker can handle a limited number of parallel browser instances (Recommended: 2 for each CPU core).
- **Sandbox:** Each execution occurs in a clean browser context to avoid cache or cookie contamination.

### 5. Enterprise Integrations (CI/CD, Webhooks)

#### Pipeline Integration (GitHub/Jenkins/GitLab)

WebFlowMaster exposes a set of RESTful APIs for triggering tests. **Example of GitHub Actions integration:**

```
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Trigger WebFlowMaster Test
        run: |
          curl -X POST "https://webflow-api.company.com/v1/execute" \
            -H "Authorization: Bearer ${ secrets.WFM_TOKEN }" \
            -d '{"planId": "prod-smoke-test"}
```

## Event-Driven Webhooks

Configure Webhooks to notify external systems (Slack, Microsoft Teams, Jira) upon critical test failure.

---

## 6. Maintenance and Disaster Recovery

---

- **Database Backups:** Daily SQL DB backups are recommended.
  - **Log Rotation:** System logs are rotated daily and kept for 30 days.
  - **Health Checks:** Constantly monitor the `/api/health` endpoint to verify the status of all services (DB, Redis, Worker).
- 

© 2026 WebFlowMaster Enterprise. Internal Use Document.